Recurrent Neural Networks (RNNs)

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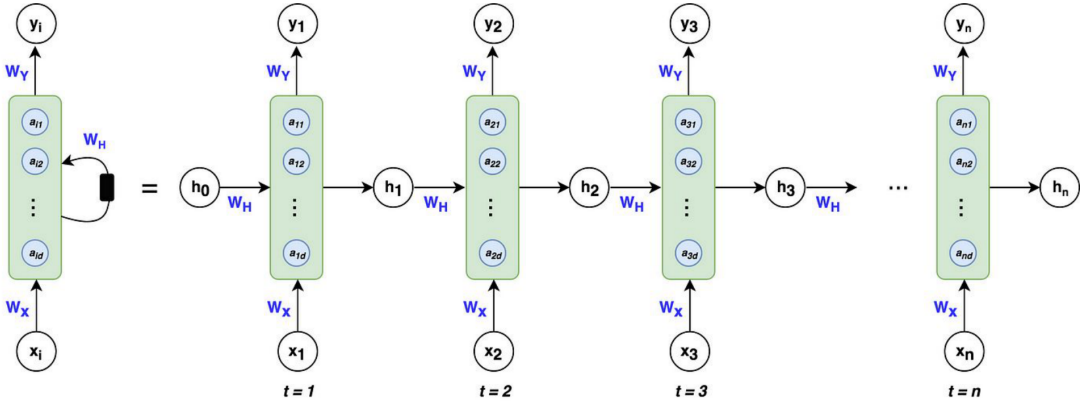


Motivation: Why Do We Need RNNs?

Why Do We Need RNNs?

- Traditional feedforward networks don't handle sequential data effectively.
- Many applications (language modeling, time-series prediction, speech recognition) require memory of past inputs.
- RNNs enable temporal dynamic behavior by maintaining hidden states.

Examples Where Order Matters:

- Translating "I am happy" vs "Happy I am"
- Predicting next stock price based on past trends
- Understanding a sentence word-by-word

Key Idea: Add a feedback loop to remember previous computations.

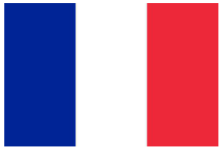
Learning Outcomes

By the end of this session, you should be able to:

- Understand the structure and working of Recurrent Neural Networks (RNNs).
- Recognize various RNN architectures (One-to-Many, Many-to-One, Many-to-Many).
- Explain the concept of shared parameters in RNNs.
- Explain why RNNs suffer from vanishing and exploding gradients.
- Understand how Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) architectures solve these problems.
- Compare LSTM and GRU architectures in terms of performance and complexity.
- Recognize limitations and future directions in sequence modeling.

RNNs : **Introduction**

Traditional Language Models



J'ai vu le match de foot

Sequence	$P(\text{Sequence})$
I saw the game of soccer	4.5 e-5
I saw the soccer game	6.0 e-5
I saw the soccer match	4.6 e-5
Saw I the game of soccer	2.6 e-9

N-grams

- Bigrams: $P(w_2|w_1) = \frac{count(w_1,w_2)}{count(w_1)}$
- Trigrams: $P(w_3|w_1,w_2) = \frac{count(w_1,w_2,w_3)}{count(w_1,w_2)}$
- $P(w_1,w_2,w_3) = P(w_1) \times P(w_2|w_1) \times P(w_3|w_2)$
- **Limitations:** Large N-grams need significant space and RAM.

N-grams: Limitations

- N-gram tables consume a lot of memory.
- **Fixed context window:** only the previous $(n-1)$ words are conditioned on.
- **Data sparsity:** larger n implies exponentially many n-grams; many combinations are rare or unseen in training data.
- **No learning or adaptability:** counts and normalization only—not learned representations in the neural sense.
- Different types of **RNNs** are the **preferred alternative**.

What is an RNN?

A neural network with loops allowing information to persist. **Core Elements:**

- Hidden State h_t captures memory of previous inputs.
- Input X_t, Output y_t.
- Parameter sharing: Same weights used across time steps.

Mathematical Formulation:

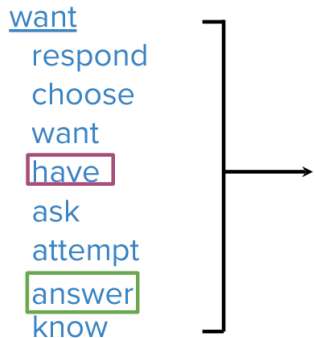
- $h_t = tanh(W_{hh}h_{t-1} + W_{xh}x_t + b_h)$
- $y_t = W_{hy}h_t + b_y$

This recurrence allows information to propagate through time.

Why Use RNNs?

- RNNs target **sequential data** where order matters (language, time series, speech).
- **Capturing context:** a hidden state carries information from previous steps.
- Example: word meaning often depends on preceding words; RNNs model such dependency.
- **Variable-length inputs** within a single architecture.
- **Temporal patterns** in language modeling, speech, financial series, etc.

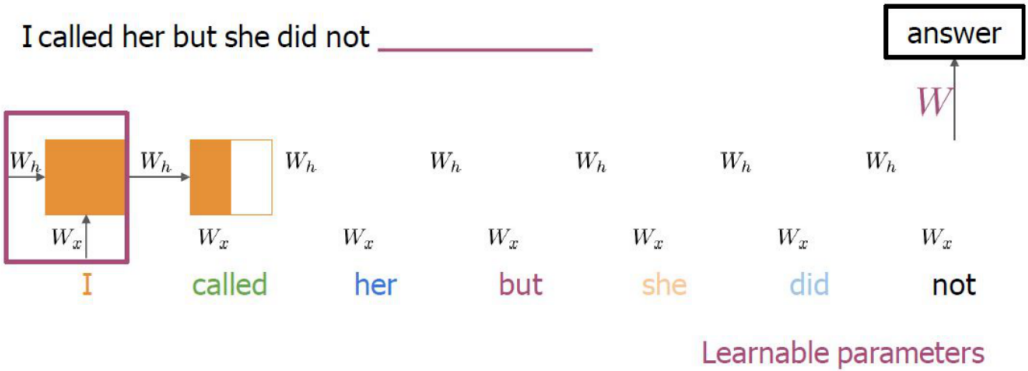
Nour was supposed to study with me. I called her but she did not answer



RNNs look at every previous word

Similar probabilities with trigram

RNNs Basic Structure



RNN : **Architecture**

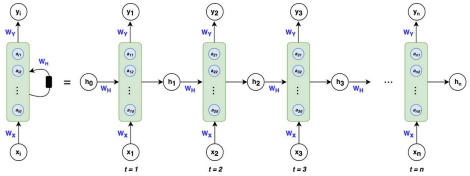
RNN Architecture: Unrolled View

Unrolled RNN:
An RNN is essentially a chain of repeating neural network modules, one for each time step.

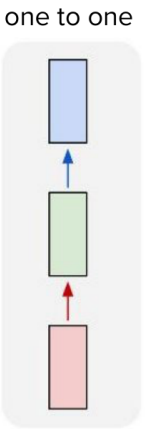
Each time step shares the same parameters:

- Input x_t
- Hidden state h_t
- Output y_t

Key Idea
Temporal representation without increasing parameter count!



Vanilla Neural Network

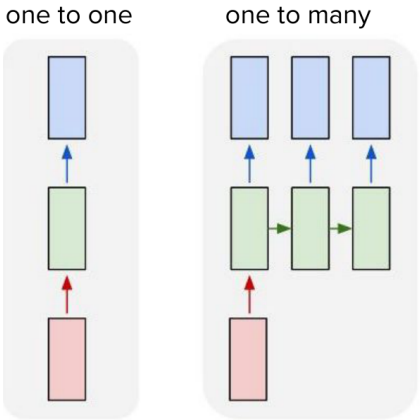


Vanilla Neural Networks

One to One


LaLiga
Santander
Real Valladolid
0
1
Real Madrid
Real Zaragoza
0
4
Real Madrid
Atletico Madrid
0
1
Real Madrid
$h^{<t_0>}$
f_W
X

RNN: Process Sequences

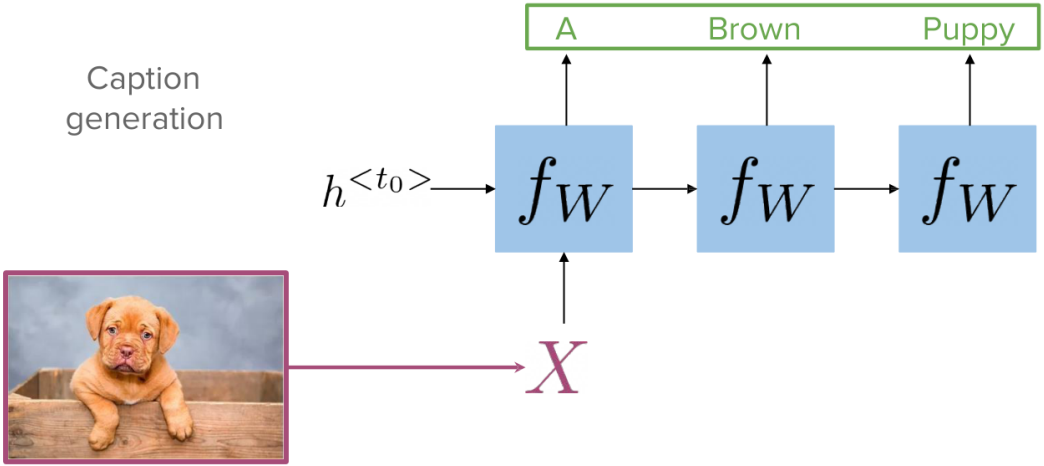


e.g. Image Captioning

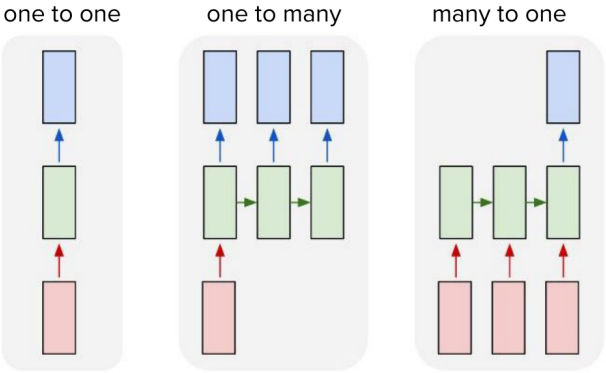
image ➜ sequence of words

One to Many

Caption generation



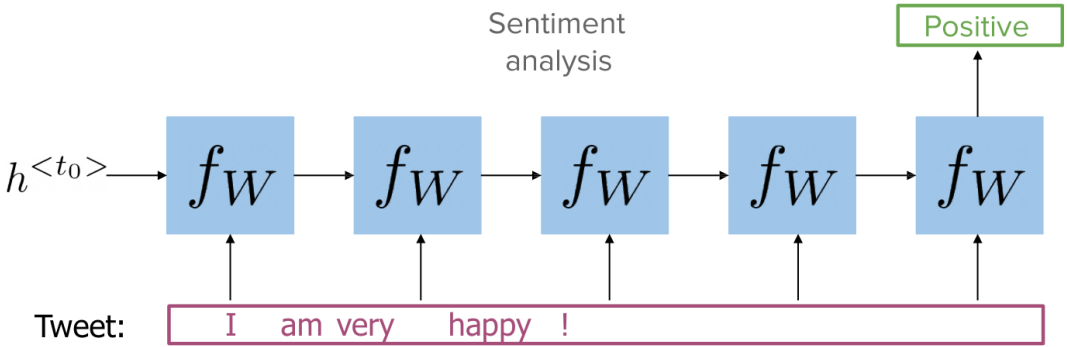
RNN: Process Sequences



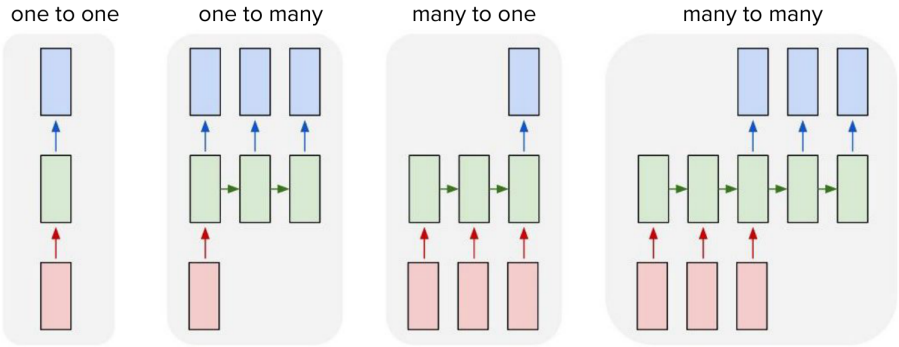
e.g. Sentiment Analysis
sequence of words ➜ sentiment

e.g. Action Prediction
sequence of video frames ➜ action class

Many to One


Sentiment analysis
Positive
$h^{<t_0>}$
fw
fw
fw
fw
fw
Tweet:
I am very happy !

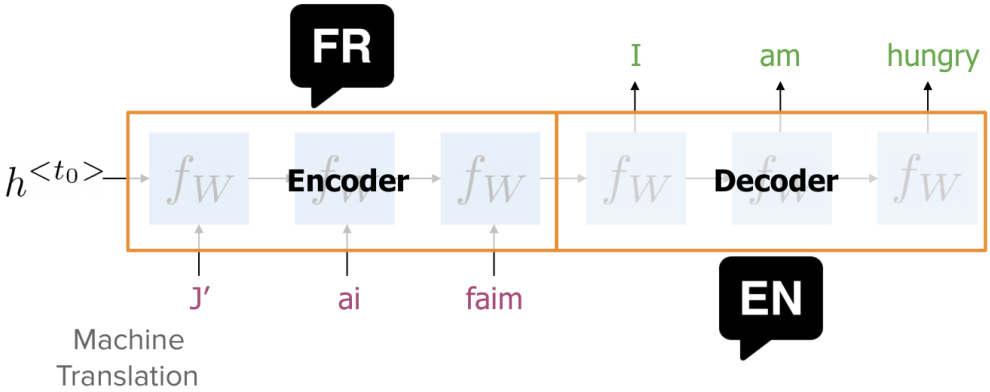
RNN: Process Sequences



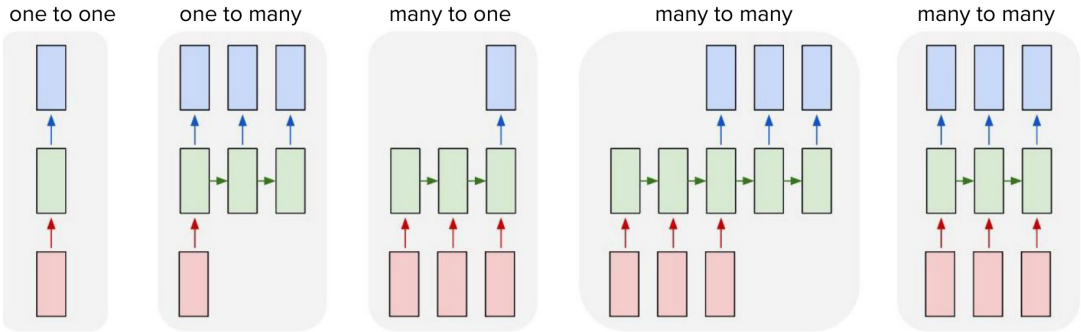
e.g. Video Captioning
sequence of video frames ➜ caption

e.g. Machine Translation
sequence of words (English) ➜ sequence of words (French)

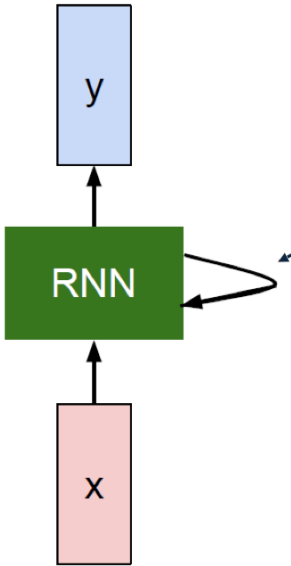
Many to Many


FR
I
am
hungry
$h^{<t_0>}$
fw
Encoder
fw
fw
fw
Decoder
fw
J'
ai
faim
EN
Machine
Translation

RNN: Process Sequences

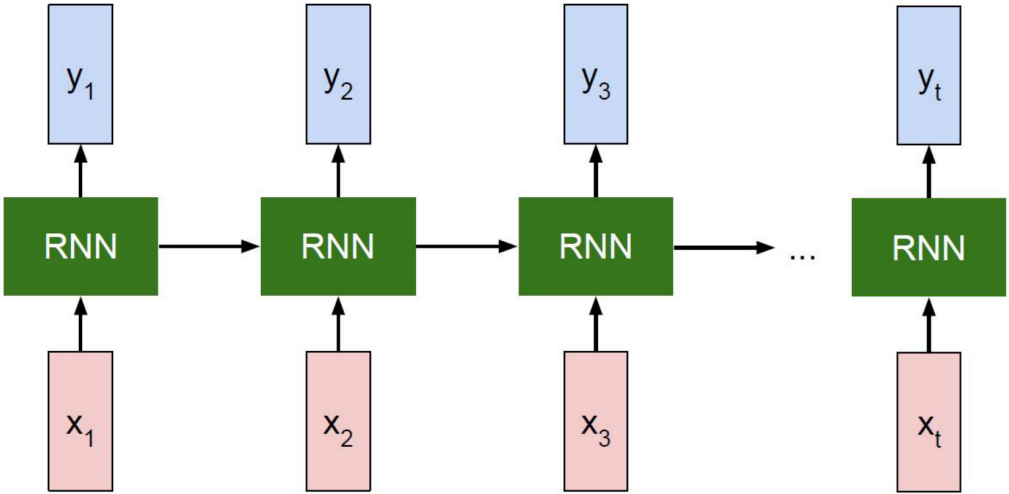


Recurrent Neural Network



Key idea: RNNs have an “internal state” that is updated as a sequence is processed

Unrolled RNN



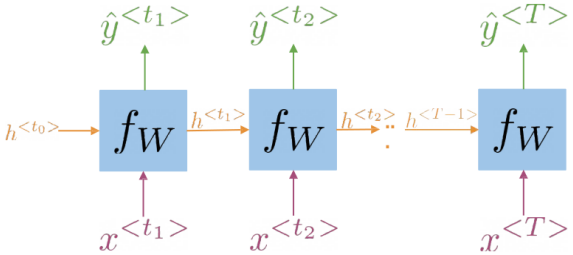
Adapted from: Stanford CS224d Lecture-8 (Fei-Fei Li, 2023) & CS231n (Zane Durante, 2025)

RNN : **Mathematical Formulations**

Math in Simple RNNs

- How RNNs propagate information (Through time!)
- How RNNs make predictions

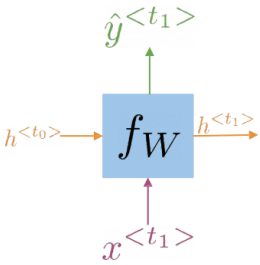
A Vanilla RNN

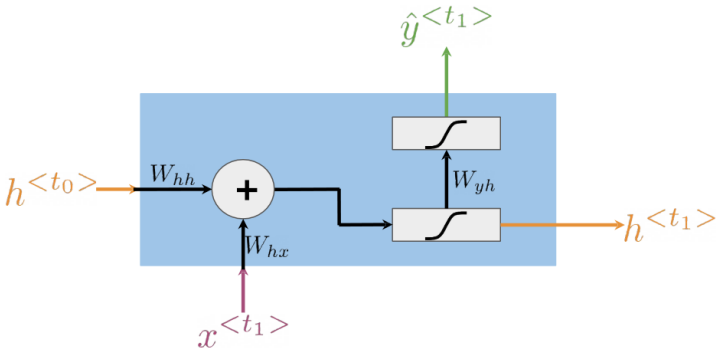


$$h^{<t>} = g(W_h[h^{<t-1>}, x^{<t>}] + b_h)$$
$$\hat{y}^{<t>} = g(W_{yh}h^{<t>} + b_y)$$

$$h^{<t>} = g(W_{hh}h^{<t-1>} + W_{hx}x^{<t>} + b_h)$$

A Vanilla RNN

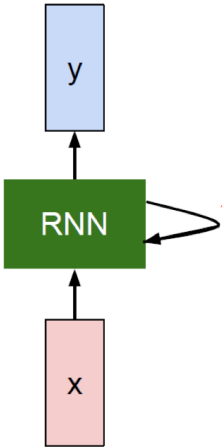




$$h^{<t>} = g(W_{hh}h^{<t-1>} + W_{hx}x^{<t>} + b_h)$$

$$\hat{y}^{<t>} = g(W_{yh}h^{<t>} + b_y)$$

A Vanilla RNN



The state consists of a single "hidden" vector h:

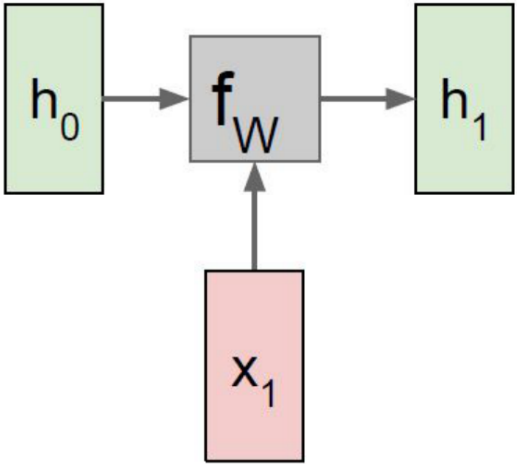
$$\mathbf{h}_t = f_W(\mathbf{h}_{t-1}, \mathbf{x}_t)$$
$$\mathbf{h}_t = \tanh(W_{hh}\mathbf{h}_{t-1} + W_{xh}\mathbf{x}_t)$$
$$\mathbf{y}_t = W_{hy}\mathbf{h}_t$$

- Sometimes called a "Vanilla RNN" or an *Elman RNN* after Prof. Jeffrey Elman.

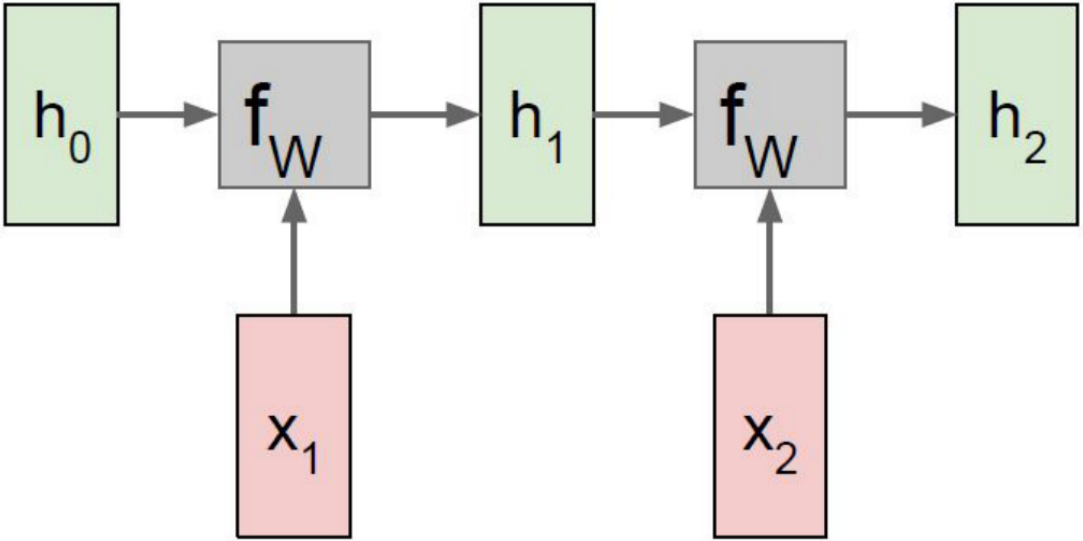
Adapted from: Stanford CS224d Lectures (R. Socher), 2023 / CSD126 (Juan Durantez, 2023)

RNN : **Parameter Sharing**

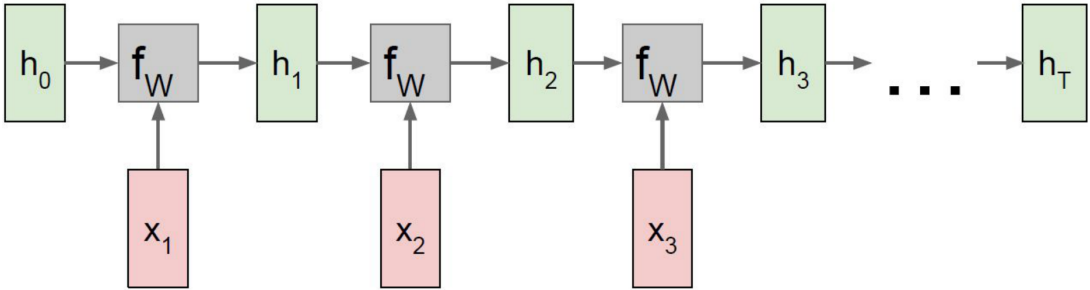
RNN: Computation Graph



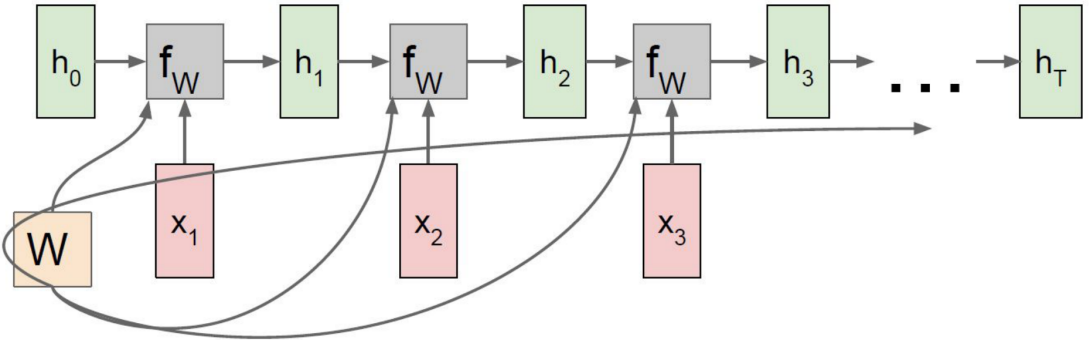
RNN: Computation Graph



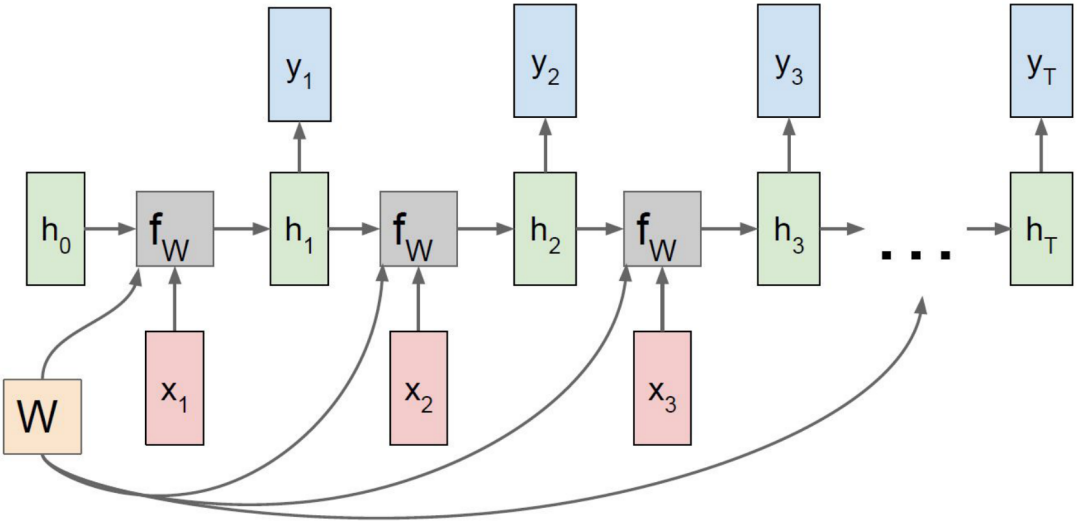
RNN: Computation Graph



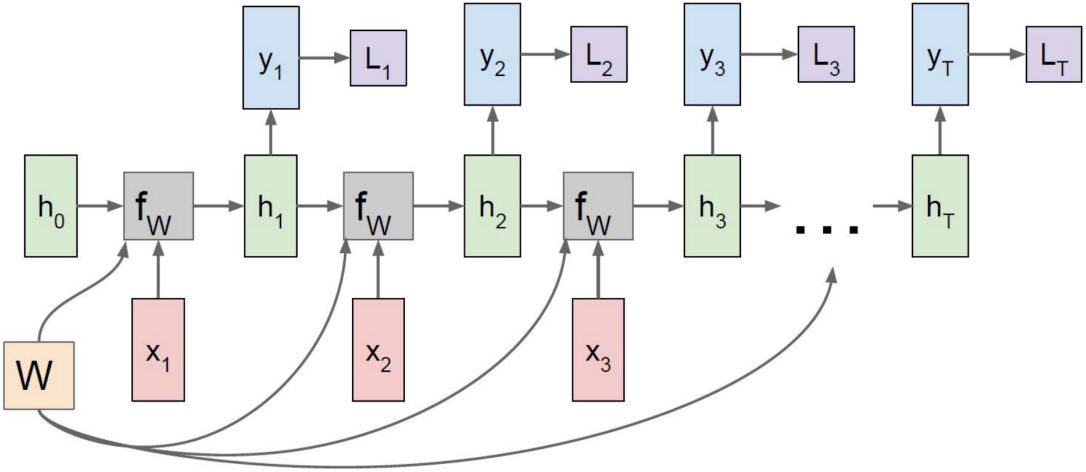
Re-use the same weight matrix at every time-step



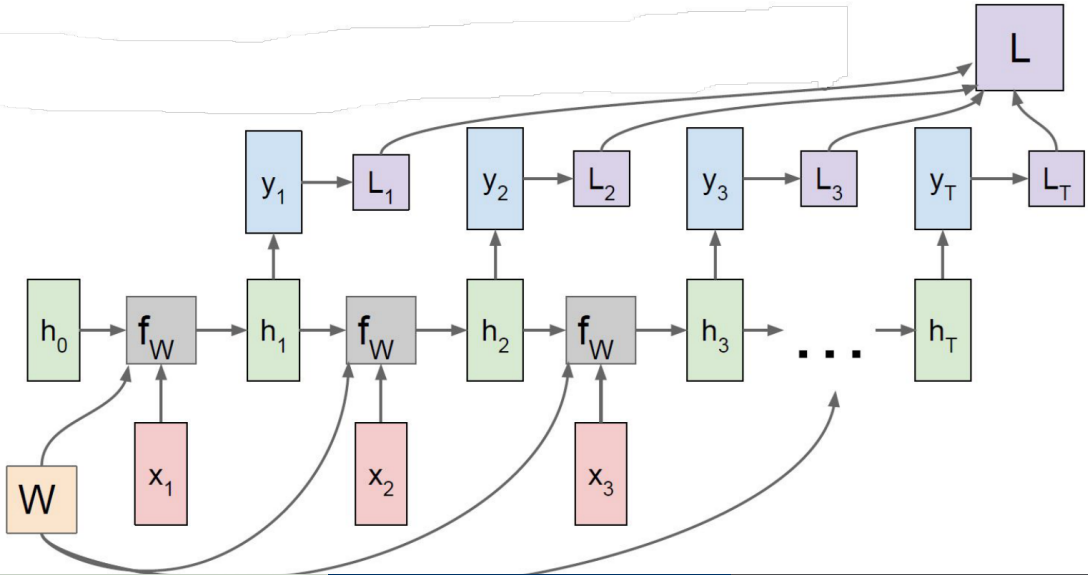
RNN Computation Graph - Many to Many



RNN Computation Graph - Many to Many

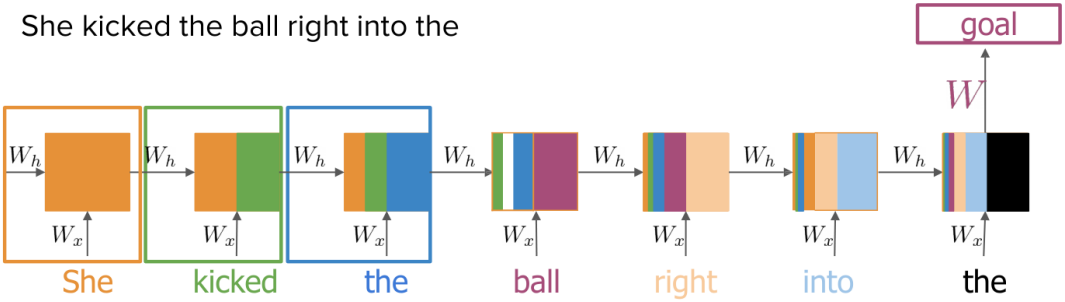


RNN Computation Graph - Many to Many



RNN : **Challenges**

RNN Basic Structure

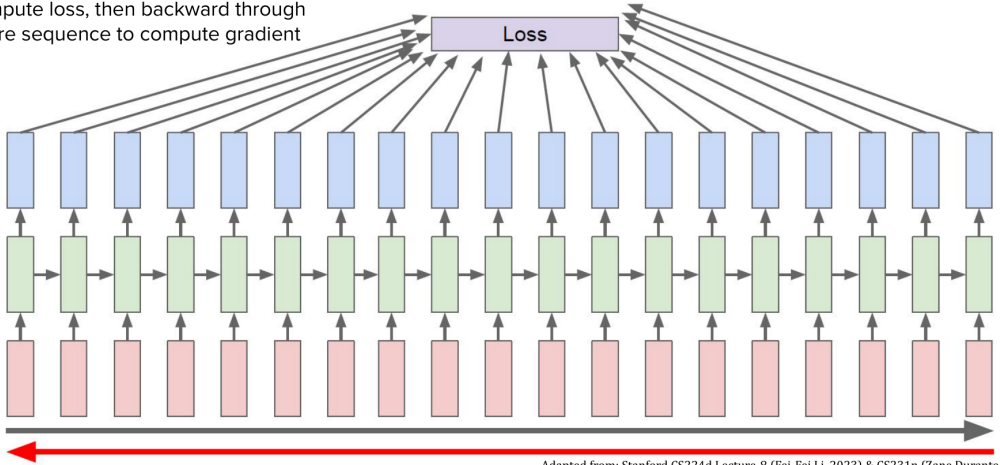


Backpropagation Through Time (BPTT)

- Training RNNs involves unfolding the network across time steps.
- Standard backpropagation is applied through this unrolled structure.

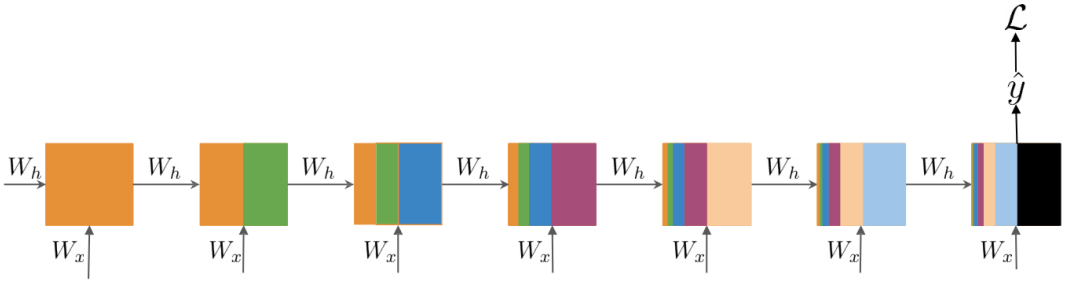
RNN: Backpropagation through time

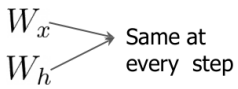
Forward through entire sequence to compute loss, then backward through entire sequence to compute gradient



Adapted from: Stanford CS224d Lecture-8 (Fei-Fei Li, 2023) & CS231n (Zane Durante, 2025)

Backpropagation through time





$$\frac{\partial L}{\partial W_h} \propto \sum_{1 \leq k \leq t} \left(\prod_{t \geq i > k} \frac{\partial h_i}{\partial h_{i-1}} \right) \frac{\partial h_k}{\partial W_h}$$

Gradient is proportional to a sum of partial derivative products

Backpropagation through time

$$\frac{\partial L}{\partial W_h} \propto \sum_{1 \leq k \leq t} \left(\prod_{t \geq i > k} \frac{\partial h_i}{\partial h_{i-1}} \right) \frac{\partial h_k}{\partial W_h}$$

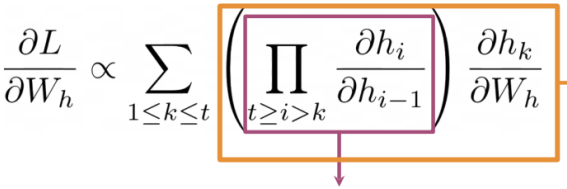
Contribution of hidden state ***k***

Length of the product proportional to how far ***k*** is from ***t***

$$\frac{\partial h_t}{\partial h_{t-1}} \frac{\partial h_{t-1}}{\partial h_{t-2}} \frac{\partial h_{t-2}}{\partial h_{t-3}} \frac{\partial h_{t-3}}{\partial h_{t-4}} \frac{\partial h_{t-4}}{\partial h_{t-5}} \frac{\partial h_{t-5}}{\partial h_{t-6}} \frac{\partial h_{t-6}}{\partial h_{t-7}} \frac{\partial h_{t-7}}{\partial h_{t-8}} \frac{\partial h_{t-8}}{\partial h_{t-9}} \frac{\partial h_{t-9}}{\partial h_{t-10}} \frac{\partial h_{t-10}}{\partial W_h}$$

Contribution of hidden state ***t-10***

Backpropagation through time



Partial derivatives <1	Contribution goes to 0	Vanishing Gradient
Partial derivatives >1	Contribution goes to infinity	**Exploding** Gradient

RNN Training Challenges

Problems

- **Vanishing Gradients:** Gradients shrink as they are propagated back, making it hard to learn long-term dependencies.
- **Exploding Gradients:** Gradients grow exponentially, leading to unstable updates.

Solutions Preview (covered in future modules)

- Use of LSTM and GRU architectures
- Gradient Clipping

Vanishing and Exploding Gradients

Backpropagation Through Time (BPTT) spreads gradients across many time steps.

Vanishing Gradients:

$$\left\| \frac{\partial \mathcal{L}}{\partial h_t} \right\| \to 0$$

- Early layers barely learn
- Forget long-term dependencies

Exploding Gradients:

$$\left\| \frac{\partial \mathcal{L}}{\partial h_t} \right\| \to \infty$$

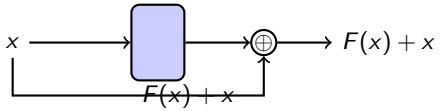
- Unstable updates, diverging weights

Solving for vanishing or exploding gradients

- Identity RNN with ReLU activation

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- Gradient clipping
- Skip connections



RNN tradeoffs

RNN Advantages

- Can process any length input.
- Computation for step t can (in theory) use information from many steps back.
- Model size doesn't increase for longer input.
- Same weights applied on every timestep, so there is symmetry in how inputs are processed.

RNN Disadvantages

- Recurrent computation is slow
- In practice, difficult to access information from many steps back

LSTM : **Long short term memory unit**

LSTM - Long Short-Term Memory

Introduced by Hochreiter & Schmidhuber (1997)

Core Idea: LSTM uses **gates** to control what to keep, forget, and output.

Key Components:

- **Forget Gate** f_t
- **Input Gate** i_t
- **Cell State** C_t
- **Output Gate** o_t

LSTM - Long Short-Term Memory

Vanilla RNN

$$h_t = \tanh\left(W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}\right)$$

LSTM

Four gates

$$\begin{pmatrix} i \\ f \\ o \\ g \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{pmatrix} W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}$$

Cell state

$$c_t = f \odot c_{t-1} + i \odot g$$

Hidden state

$$h_t = o \odot \tanh(c_t)$$

Hochreiter and Schmidhuber, "Long Short Term Memory", Neural Computation 1997

LSTM - Long Short-Term Memory: Equations

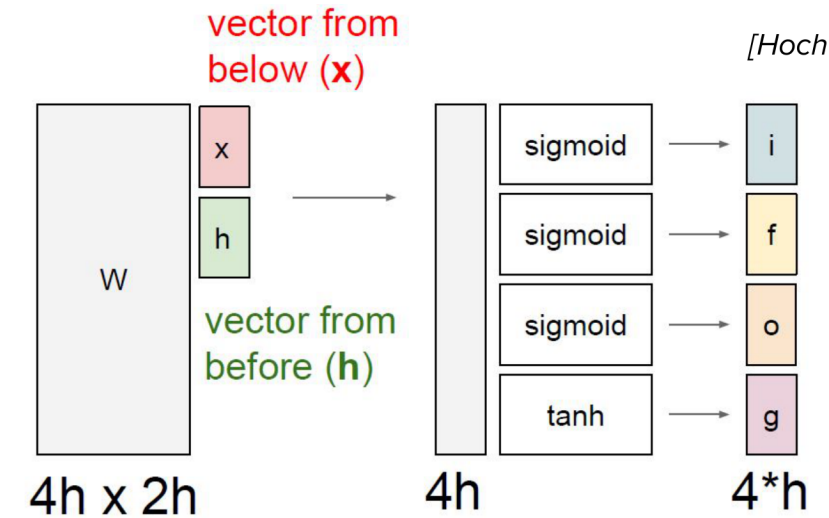
Equations:

$$
\begin{aligned}
f_t &= \sigma\left(W_f[x_t, h_{t-1}] + b_f\right) \\
i_t &= \sigma\left(W_i[x_t, h_{t-1}] + b_i\right) \\
\tilde{C}_t &= \tanh\left(W_c[x_t, h_{t-1}] + b_c\right) \\
C_t &= f_t * C_{t-1} + i_t * \tilde{C}_t \\
o_t &= \sigma\left(W_o[x_t, h_{t-1}] + b_o\right) \\
h_t &= o_t * \tanh\left(C_t\right)
\end{aligned}
$$

Helps retain long-term dependencies

LSTMs - A Memorable Solution

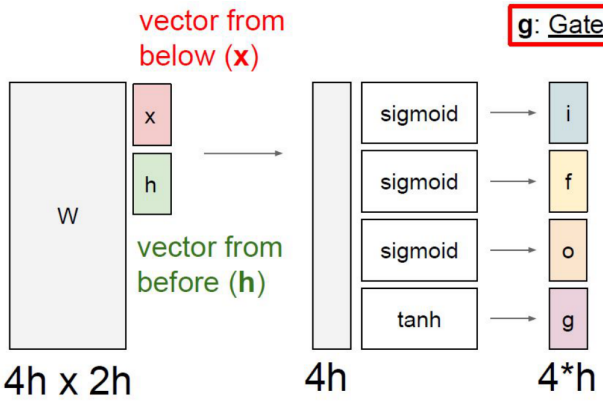
- Learns when to remember and when to forget
- **Basic anatomy:**
  - A cell state
  - A hidden state
  - Multiple gates
- Gates allow gradients to avoid vanishing and exploding



[Hochreiter et al., 1997]

Adapted from: Stanford CS224d Lecture-8 (Fei-Fei Li, 2023) & CS231n (Zane Durante, 2025)

[Hochreiter et al., 1997]



g: Gate gate (?), How much to write to cell

$$\begin{pmatrix} i \\ f \\ o \\ g \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{pmatrix} W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}$$

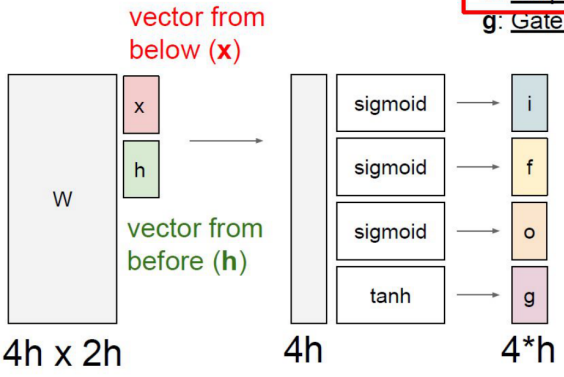
$$c_t = f \odot c_{t-1} + i \odot g$$

$$h_t = o \odot \tanh(c_t)$$

Adapted from: Stanford CS224d Lecture-8 (Fei-Fei Li, 2023) & CS231n (Zane Durante, 2025)

[Hochreiter et al., 1997]

i: Input gate, whether to write to cell
f: Forget gate, Whether to erase cell
o: Output gate, How much to reveal cell
g: Gate gate (?), How much to write to cell



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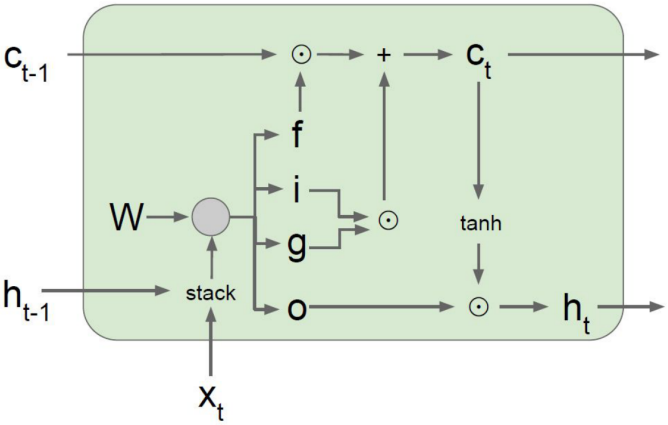
Long Short-Term Memory

[Hochreiter et al., 1997]



$$\begin{pmatrix} i \\ f \\ o \\ g \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{pmatrix} W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}$$

$$c_t = f \odot c_{t-1} + i \odot g$$

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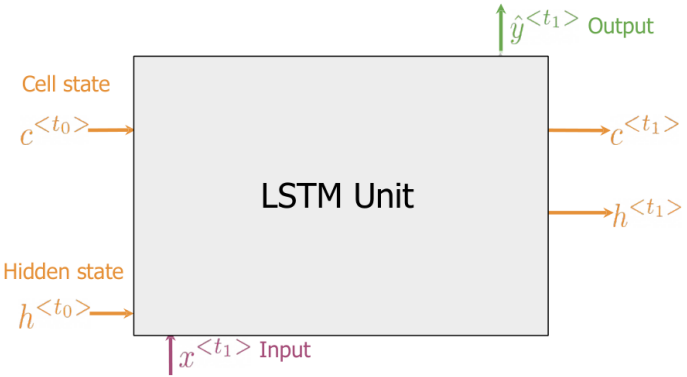
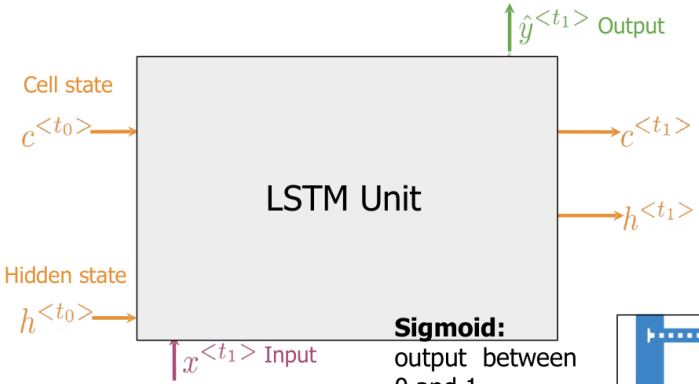
Gates in LSTM



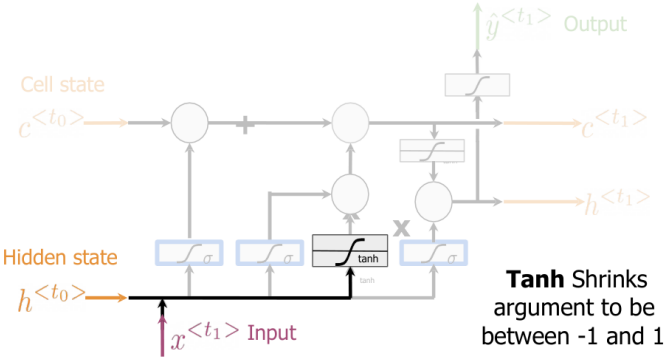
1. **Forget Gate**: information that is no longer important
2. **Input Gate**: information to be stored
3. **Output Gate**: information to use at current step

Gates in LSTM



1. **Forget Gate**: information that is no longer important
2. **Input Gate**: information to be stored
3. **Output Gate**: information to use at current step

Candidate Cell State

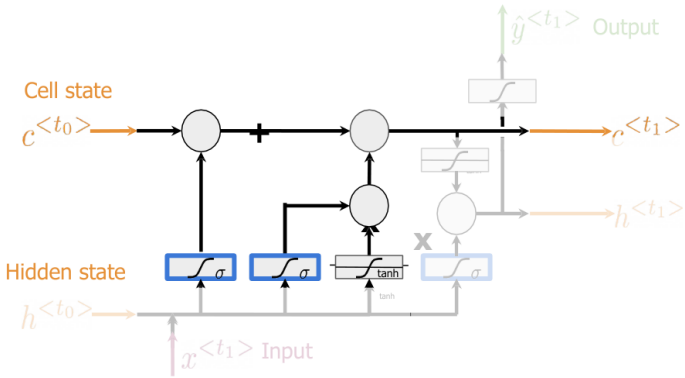


Candidate cell state

Information from the previous **hidden state** and current **input**

Tanh Shrinks argument to be between -1 and 1 → **Other activations could be used**

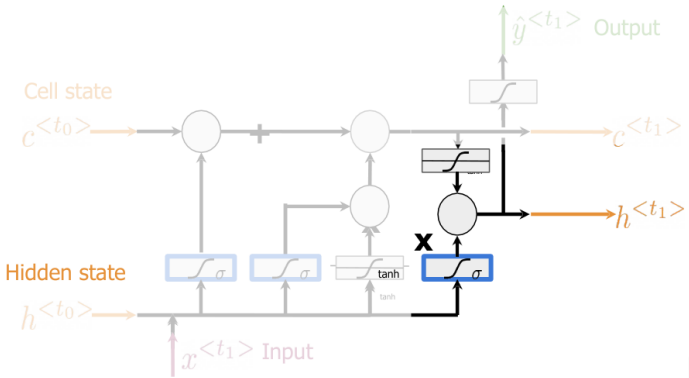
New Cell State



New Cell state

Add information from the **candidate cell state** using the **forget** and **input gates**

New Hidden State

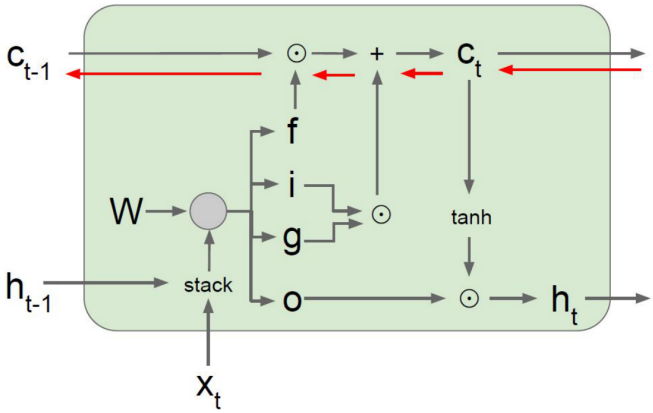


New Hidden State

Select information from the **new cell state** using the **output gate**

The **Tanh** activation could be omitted

LSTM: Gradient Flow



[Hochreiter et al., 1997]

Backpropagation from c_t to c_{t-1} only elementwise multiplication by f, no matrix multiply by W

$$\begin{pmatrix} i \\ f \\ o \\ g \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{pmatrix} W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}$$

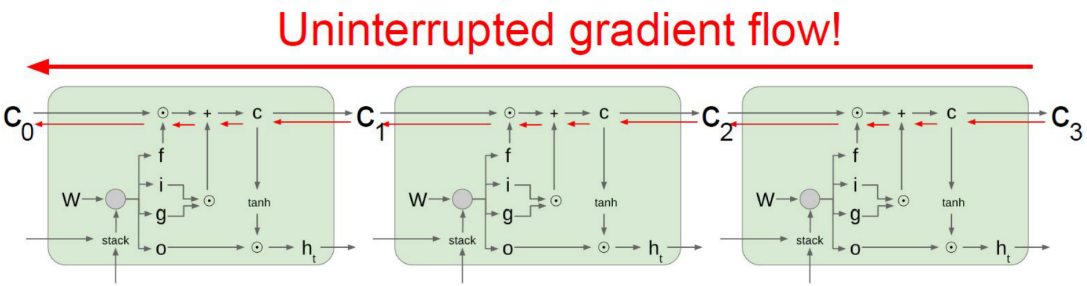
$$c_t = f \odot c_{t-1} + i \odot g$$

$$h_t = o \odot \tanh(c_t)$$

Adapted from: Stanford CS224d Lecture-8 (Fei-Fei Li, 2023) & CS231n (Zane Durante, 2025)

LSTM: Gradient Flow

[Hochreiter et al., 1997]



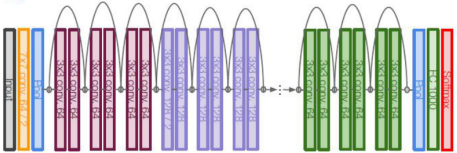
Adapted from: Stanford CS224d Lecture-8 (Fei-Fei Li, 2023) & CS231n (Zane Durante, 2025)

LSTM: Gradient Flow

[Hochreiter et al., 1997]

Uninterrupted gradient flow!

Similar to ResNet!



In between:

Highway Networks

$$g = T(x, W_T)$$

$$y = g \odot H(x, W_H) + (1 - g) \odot x$$

Srivastava et al, "Highway Networks", ICML DL Workshop 2015

Adapted from: Stanford CS224d Lecture-8 (Fei-Fei Li, 2023) & CS231n (Zane Durante, 2025)

Do LSTMs solve the vanishing gradient problem?

- The LSTM architecture makes it easier for the RNN to preserve information over many timesteps.
  - e.g. if the forget gate $f = 1$ and input gate $i = 0$, then the information of that cell is preserved indefinitely.
  - By contrast, it is harder for a vanilla RNN to learn a recurrent weight matrix W_h that preserves information in the hidden state.
- LSTM **doesn't guarantee** that there is no vanishing/exploding gradient, but it does provide an easier way for the model to learn long-distance dependencies.

GRU : **Gated Recurrent Unit**

GRU - Gated Recurrent Unit

Introduced by Cho et al., 2014

Core Idea: GRU uses **gates** to control what to keep, forget, and output.

Key Components:

- **Update Gate** z_t
- **Reset Gate** r_t

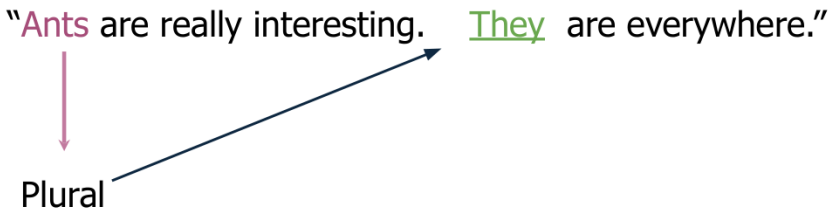
GRU - Gated Recurrent Unit: Equations

Equations:

$$
\begin{aligned}
z_t &= \sigma(W_z[x_t, h_{t-1}] + b_z) \\
r_t &= \sigma(W_r[x_t, h_{t-1}] + b_r) \\
\tilde{h}_t &= \tanh(W_h[x_t, r_t * h_{t-1}] + b_h) \\
h_t &= (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t
\end{aligned}
$$

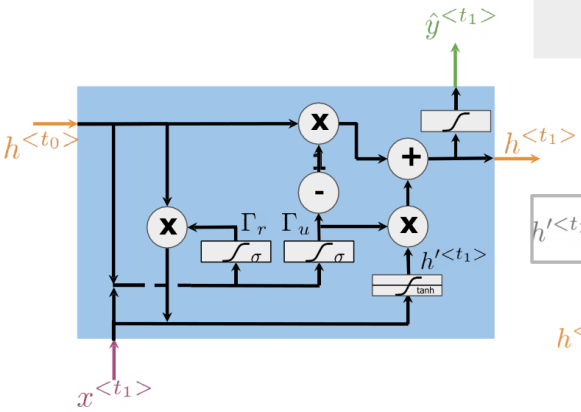
Comparable performance to LSTM
Faster training, fewer parameters

GRU - Gated Recurrent Units



Relevance and update gates to remember important prior information

GRU - Gated Recurrent Unit



Gates to keep/update relevant information in the hidden state

$$\Gamma_r = \sigma(W_r[h^{<t_0>}, x^{<t_1>}] + b_r)$$
$$\Gamma_u = \sigma(W_u[h^{<t_0>}, x^{<t_1>}] + b_u)$$

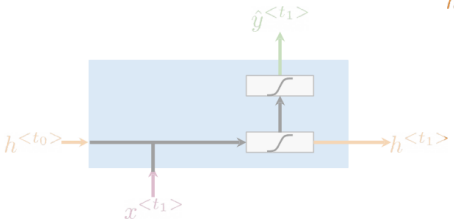
$$h'^{<t_1>} = \tanh\left(W_h[\Gamma_r * h^{<t_0>}, x^{<t_1>}] + b_h\right)$$

Hidden state candidate

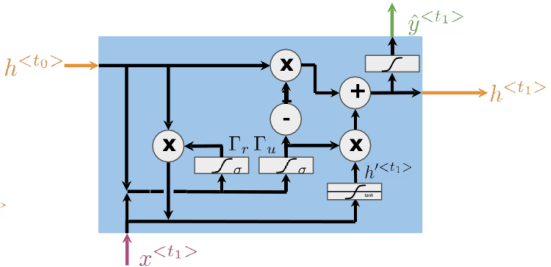
$$h^{<t_1>} = (1 - \Gamma_u) * h^{<t_0>} + \Gamma_u * h'^{<t_1>}$$

$$\hat{y}^{<t_1>} = g(W_y h^{<t_1>} + b_y)$$

Vanilla RNN vs GRUs



$$h^{<t>} = g(W_h[h^{<t-1>}, x^{<t>}] + b_h)$$
$$\hat{y}^{<t>} = g(W_{yh}h^{<t>} + b_y)$$



$$\Gamma_u = \sigma(W_u[h^{<t_0>}, x^{<t_1>}] + b_u)$$
$$\Gamma_r = \sigma(W_r[h^{<t_0>}, x^{<t_1>}] + b_r)$$
$$h'^{<t_1>} = \tanh(W_h[\Gamma_r * h^{<t_0>}, x^{<t_1>}] + b_h)$$
$$h^{<t_1>} = (1 - \Gamma_u) * h^{<t_0>} + \Gamma_u * h'^{<t_1>}$$
$$\hat{y}^{<t_1>} = g(W_y h^{<t_1>} + b_y)$$

LSTM vs GRU

Feature	LSTM	GRU
Gates	3 (i, f, o)	2 (z, r)
Memory Cell	Yes	No
Complexity	Higher	Lower
Training Speed	Slower	Faster
Performance	Great for long sequences	Similar or better on short tasks

Tip: Try GRU first for faster results; switch to LSTM if performance suffers.

Use Cases for GRU/LSTM

NLP Tasks:

- Language Modeling
- Machine Translation
- Sentiment Analysis
- Chatbots

Audio & Time Series:

- Music Generation
- Speech Recognition
- Anomaly Detection

Video & Sequential Vision:

- Action Recognition
- Video Captioning

Limitations of GRU/LSTM

- Sequential computation — hard to parallelize
- Forget long-term dependencies
- Slow training due to sequential nature
- Struggle with varying-length sequences

Future Directions

- **Transformers:** Fully parallelized sequence modeling using attention
- **Efficient Attention:** Longformer, Linformer, etc. for long sequences
- **Neural Memory Networks:** Explicit memory read/write
- **Recurrent Attention Models**
- **Hybrid Architectures:** RNN + CNN + Attention

RNNs are still used in edge devices for efficient modeling

Summary

- RNNs struggle with long dependencies due to vanishing/exploding gradients
- GRU and LSTM improve memory retention using gating mechanisms
- GRU is simpler and faster; LSTM is more expressive
- Attention and transformers now dominate, but RNNs remain relevant in many domains

References

These slides have been adapted from

- Younes Mourri & Lukasz Kaiser, Natural Language Processcing Specialization, DeepLearning.AI

Foundational Papers:

- Hochreiter, S., & Schmidhuber, J. (1997). *Long Short-Term Memory*. Neural Computation.
- Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., & Bengio, Y. (2014). *Learning Phrase Representations using RNN Encoder–Decoder with GRU*. EMNLP.
- Pascanu, R., Mikolov, T., & Bengio, Y. (2013). *On the difficulty of training RNNs*. ICML.
- Bengio, Y., Simard, P., & Frasconi, P. (1994). *Learning long-term dependencies*. IEEE Transactions on Neural Networks.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). *Attention Is All You Need*. NeurIPS.

References

Resources:

- Karpathy's RNN Blog: https://karpathy.github.io/2015/05/21/rnn-effectiveness/
- CS231n Lecture Notes on RNNs and LSTM
- DeepLearning.ai NLP Specialization — Coursera
- MIT 6.S191 Deep Learning Lecture Slides